CS 4412 Data Mining

Final Report: Pattern Discovery in the Global Terrorism Database

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Introduction

This project analyzes the Global Terrorism Database to identify meaningful patterns in terrorist incidents from 2000 onward. The goal of this analysis is not to predict future events, but to uncover structure within the data and better understand how incidents differ across multiple dimensions. By focusing on pattern discovery, the analysis emphasizes interpretation and insight rather than predictive accuracy.

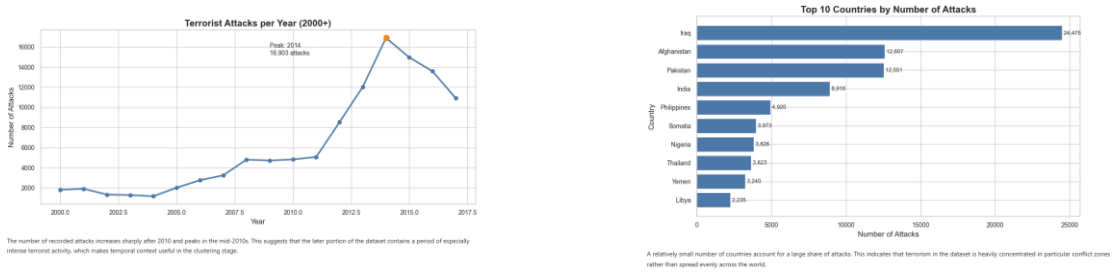
The dataset contains records of incidents worldwide, including features such as attack type, weapon type, target type, geographic region, and casualty counts. Restricting the dataset to more recent years improves consistency and reduces issues related to incomplete or noisy historical data. This allows the analysis to focus on clearer and more comparable patterns.

The study is guided by three discovery questions. First, whether terrorist incidents form natural clusters based on their characteristics. Second, how geographic and temporal context influences those clusters. Third, whether alternative clustering approaches reveal additional structure not captured by traditional methods.

Understanding these patterns provides insight into how different forms of violent activity manifest across regions and time. The objective is to move beyond simple categorization and develop a deeper understanding of how incidents relate to one another within a complex global context.

Methods

The analysis begins with exploratory data analysis to understand the structure and distribution of the dataset. Trends in attack frequency, geographic concentration, and casualty severity are examined to identify key patterns. Because casualty data is highly skewed, a logarithmic transformation is applied to reduce the influence of extreme values and improve comparability across incidents.



Attacks over time *Top countries by attack count*

Figure 1. Exploratory context: attack frequency over time and geographic concentration.

Categorical variables such as attack type, target type, weapon type, and region are encoded using one-hot encoding. This approach preserves categorical distinctions without introducing artificial

ordering, which is important for clustering algorithms that rely on distance calculations. All features are then standardized to ensure that no single variable disproportionately influences the clustering process.

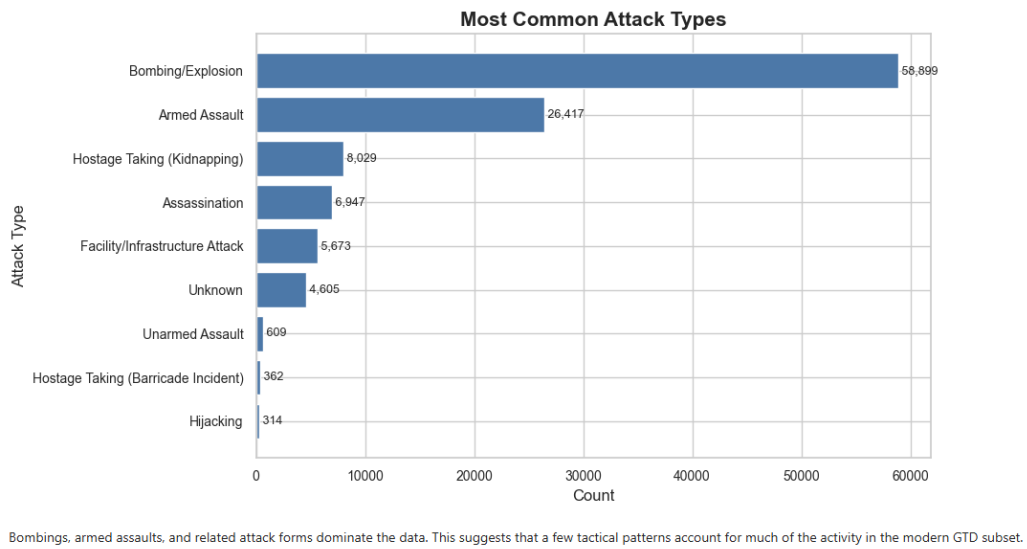
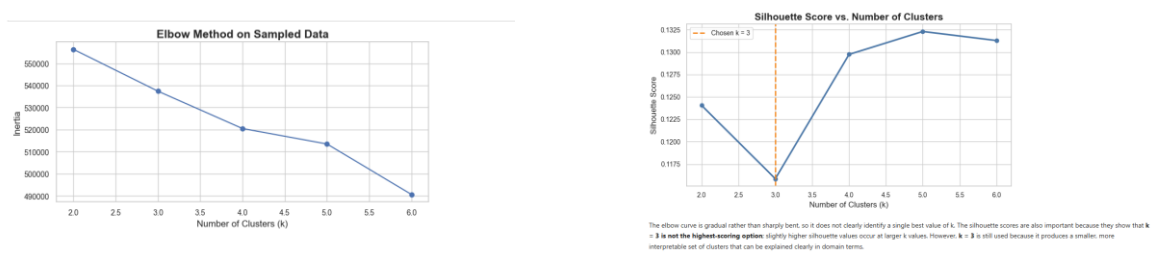


Figure 2. Most common attack types in the filtered dataset.

K-Means clustering is used as the primary method for grouping incidents. The elbow method and silhouette scores are used to evaluate different values of k. Although silhouette scores do not favor k equals three as the optimal value, this value is selected because it produces clusters that are more interpretable and meaningful in the context of the data.



Elbow method *Silhouette score curve*

Figure 3. K-Means model selection diagnostics. The selected k = 3 is interpretability-driven rather than the highest silhouette score.

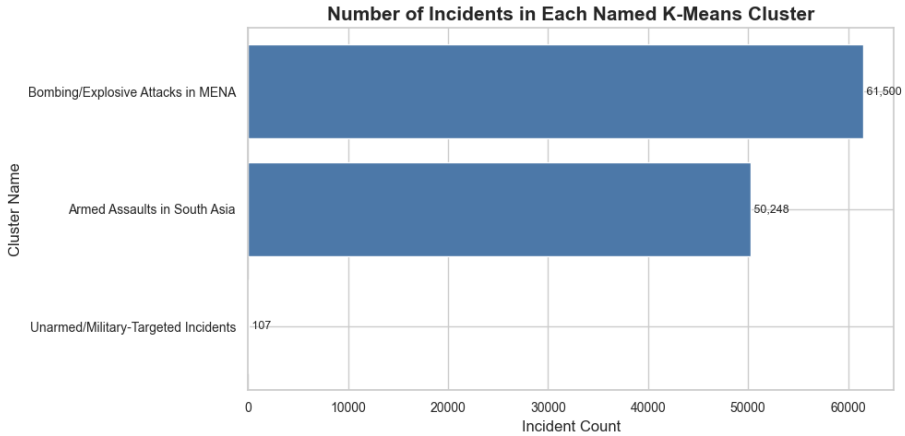
Principal Component Analysis is used to reduce dimensionality for visualization. However, because the first two components capture only a small portion of the total variance, PCA is treated as a visualization tool rather than definitive evidence of cluster separation.

DBSCAN is applied as an alternative clustering method to explore density-based structure. A parameter sweep is conducted to evaluate different configurations, revealing that the dataset does not naturally form a small number of dense clusters. Finally, a decision tree is used as an interpretability tool to identify which features most strongly differentiate the clusters.

Results

K-Means clustering produces three distinct groups of incidents. Each cluster exhibits unique characteristics in terms of attack type, severity, and geographic distribution. The first cluster is

dominated by bombing and explosive attacks, often associated with higher casualty counts and concentration in regions such as the Middle East and North Africa.



The cluster counts show how incidents are distributed across the three broad groups. This gives an immediate sense of whether the segmentation is balanced or dominated by one large cluster.

Figure 4. Distribution of incidents across the named K-Means clusters.

The second cluster consists of armed assaults, particularly in South Asia. These incidents are more frequent but generally produce moderate casualty levels. This suggests a pattern of ongoing localized violence rather than isolated high-impact events.

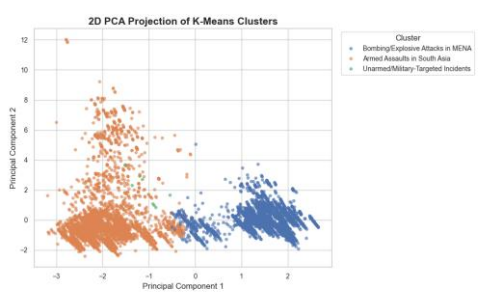
The third cluster includes incidents that involve military targets or lower-severity interactions. These may reflect more complex forms of conflict that do not fit neatly into traditional definitions of terrorism. The presence of military targets suggests overlap with broader conflict dynamics.

cluster	top_attacktype1_txt	top_targtype1_txt	top_weaptype1_txt	top_region_txt
0	Bombing/Explosion	Private Citizens & Property	Explosives	Middle East & North Africa
1	Unarmed Assault	Military	Vehicle (not to include vehicle-borne explosiv...	Middle East & North Africa
2	Armed Assault	Private Citizens & Property	Firearms	South Asia

The most common categories in each cluster make the cluster patterns easier to interpret in domain terms. Instead of thinking only in terms of cluster numbers, this table highlights the attack types, targets, weapons, and regions that characterize each group most clearly.

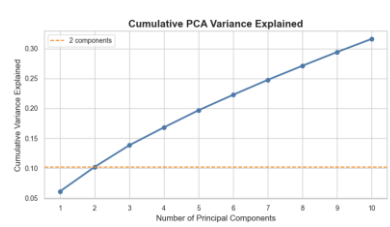
Figure 5. Dominant attack, target, weapon, and region categories by cluster.

The PCA visualization shows partial separation between clusters but captures only a limited portion of total variance. As a result, the visualization provides only a rough summary of the data structure. The true clustering exists in a higher-dimensional feature space.



The 2D PCA projection is useful as a rough visual summary, but it should not be treated as strong evidence of cluster separation by itself. The first two principal components capture only a limited portion of the full feature space, so the actual K-Means clusters are defined in the original high-dimensional standardized feature matrix, not only in this 2D plot.

2D PCA projection



This plot makes the PCA limitation visible. The first two components are useful for a compact visual sketch, but they explain only a small portion of the total variance. The higher-dimensional clustering results should therefore be interpreted using the full feature space and the cluster profile tables, not the 2D plot alone.

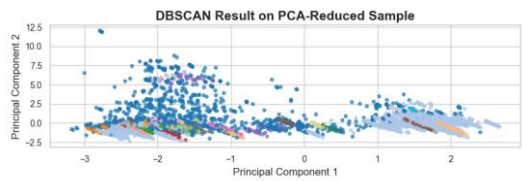
Cumulative PCA variance

Figure 6. PCA visualization and explained variance. The 2D projection is useful as a sketch but does not represent the full feature space.

DBSCAN produces a large number of small clusters and identifies a portion of points as noise. This indicates that the dataset does not exhibit strong global density-based clustering. Instead, it contains many localized patterns that do not combine into a small number of cohesive groups.

	eps	min_samples	clusters_found	noise_percent
0	1.2	10	93	17.34
1	1.2	20	62	24.62
2	1.5	10	53	12.49
3	1.5	20	36	18.39
4	1.8	10	18	7.48
5	1.8	20	10	10.98

DBSCAN parameter sweep



DBSCAN projection

Figure 7. DBSCAN results show fragmentation into many small density-based groups instead of a few broad clusters.

The decision tree analysis highlights key features that distinguish clusters, including casualty levels, attack type indicators, and regional variables. This confirms that the clusters are based on meaningful differences rather than arbitrary groupings.

Interpretation

The clustering results provide a framework for understanding how terrorist incidents differ not only in method but also in intent, context, and operational structure. Rather than representing isolated categories, the clusters reflect broader behavioral patterns shaped by both tactical decisions and environmental constraints.

The bombing-dominated cluster can be interpreted as representing highly strategic forms of violence. These incidents often involve planning, coordination, and resource allocation that exceed what is required for more opportunistic attacks. The use of explosives allows attackers to extend their impact beyond immediate physical proximity, increasing both casualties and psychological effect. This aligns with widely recognized characteristics of terrorism that emphasize visibility and disruption.

At the same time, this cluster reflects the importance of regional infrastructure and access to materials. The repeated presence of these attacks in certain regions suggests that the ability to carry out high-impact operations is closely tied to local conditions, including supply chains, organizational networks, and prolonged instability. These factors collectively shape the tactical landscape in which such attacks occur.

In contrast, the armed assault cluster reflects a different operational model. These incidents tend to occur more frequently and require fewer resources to execute. The reliance on firearms indicates a shift toward direct engagement rather than indirect or large-scale methods. This suggests that these attacks may be driven by immediacy and accessibility rather than long-term strategic planning.

The frequency of these incidents is particularly important. While individual events may not produce large casualty counts, their cumulative effect can contribute significantly to instability. This pattern

indicates that sustained, lower-intensity violence can be just as impactful as isolated high-severity events when considered over time.

The third cluster introduces a more nuanced interpretation. The presence of military targets and relatively lower severity suggests that these incidents may not always align with conventional definitions of terrorism. Instead, they may represent interactions within broader conflict systems, including engagements between non-state actors and organized forces.

This raises important questions about classification. In regions experiencing ongoing conflict, the distinction between terrorism and other forms of violence becomes less clear. Some events may be labeled as terrorism based on dataset criteria, but in practice they may resemble insurgent activity, militia operations, or tactical engagements within a larger conflict.

This ambiguity is especially relevant in regions such as Iraq during the mid-2010s. During this period, multiple actors, including militias, insurgent groups, and formal military forces, were engaged in complex interactions. As a result, some incidents classified as terrorism may reflect broader conflict dynamics rather than distinct acts of terrorism.

This insight has important implications for interpreting the clusters. Rather than viewing them strictly as categories of terrorism, it may be more accurate to interpret them as patterns of violent activity within different operational contexts. This perspective acknowledges the limitations of categorical labels and emphasizes the importance of context.

The DBSCAN results further reinforce this interpretation. The fragmentation of the dataset into many small clusters suggests that the data does not naturally group into a small number of dense regions. Instead, it reflects a wide variety of localized patterns that differ across regions and conditions.

This indicates that terrorist activity, as represented in the dataset, is not governed by a single underlying structure. Instead, it emerges from a combination of factors, including geography, political conditions, resource availability, and organizational strategy. These factors interact in complex ways, producing patterns that cannot be fully captured by a single clustering approach.

Overall, the interpretation of these results highlights the importance of combining statistical analysis with contextual understanding. While clustering provides a useful way to organize the data, the meaning of those clusters depends on how they are interpreted within real-world conditions. This reinforces the idea that data-driven insights must be evaluated alongside domain knowledge to produce meaningful conclusions.

Critical Assessment

The analysis is subject to several methodological and conceptual limitations that affect how the results should be interpreted. One of the primary limitations is the use of K-Means clustering, which assumes that clusters are relatively compact and separable. In practice, the structure of the data may not conform to these assumptions, which can influence the resulting groupings.

The silhouette scores provide additional insight into this limitation. Although the selected value of k provides interpretable clusters, it does not maximize separation. This indicates that the clustering solution represents a tradeoff between statistical optimization and interpretability. While this choice is justified for analytical clarity, it introduces subjectivity into the modeling process.

	k	silhouette_score
0	2	0.124055
1	3	0.115870
2	4	0.129741
3	5	0.132302
4	6	0.131290

Figure 8. Silhouette score table showing that k = 3 was not the highest-scoring option.

Another limitation arises from the high-dimensional feature space created through one-hot encoding. While this approach preserves categorical distinctions, it increases sparsity and can distort distance-based calculations. As a result, the clustering algorithm may be influenced by the structure of the encoding rather than purely by underlying patterns in the data.

PCA is used to visualize the data, but the first two components capture only a small fraction of the total variance. This means that the visual representation of clusters is incomplete and should not be used as primary evidence of separation. The true clustering structure exists in a higher-dimensional space that cannot be easily visualized.

The DBSCAN results highlight another important limitation. The method is highly sensitive to parameter selection, and in this case it does not produce a small number of clear clusters. Instead, it reveals fragmentation, suggesting that the dataset lacks strong global density structure. While this is informative, it also indicates that the data does not conform well to density-based assumptions.

Ethical considerations are also central to this analysis. The dataset represents real-world incidents involving violence, and reducing these events to numerical patterns risks oversimplification. Care must be taken to ensure that analysis does not remove the human context behind the data.

In addition, the classification of incidents introduces potential ambiguity. The definition of terrorism used in the dataset may include events that overlap with other forms of violence, particularly in regions experiencing ongoing conflict. This means that the patterns identified in the clusters may reflect broader conflict dynamics rather than purely terrorist behavior.

This issue is particularly relevant in cases where multiple actors operate within the same environment. The presence of militias, insurgent groups, and formal military forces can blur distinctions between different types of events. As a result, the dataset may group together incidents that differ significantly in intent and context.

These limitations do not invalidate the results, but they emphasize the importance of careful interpretation. The findings should be viewed as patterns within the dataset rather than definitive statements about real-world phenomena. Recognizing these constraints is essential for responsible analysis and meaningful conclusions.

Conclusion

This analysis demonstrates that meaningful patterns can be identified in the Global Terrorism Database through clustering techniques. The results reveal distinct forms of violent activity that differ in tactics, severity, and geographic context.

At the same time, the findings must be interpreted carefully due to methodological limitations and ambiguity in classification. The patterns identified reflect how the dataset organizes incidents rather than definitive categorizations of real-world events.

Future work could explore additional features, alternative clustering methods, or more detailed regional analysis to further refine these insights. Overall, this project highlights both the value and the limitations of data mining when applied to complex and sensitive real-world datasets.